

## Algorithms – Exercise Sheet 5

Discussion on Monday, 2026-06-29

### Exercise 1 (Huffman's Algorithm and Coding Trees)

The following table gives the relative frequencies of the letters A, B, C, ..., J in an unknown language. (The entries have already been sorted by frequency.)

| $a_i$ | J    | B    | G    | F    | C    | D    | H    | I    | A    | E    |    |    |    |    |    |    |    |    |    |
|-------|------|------|------|------|------|------|------|------|------|------|----|----|----|----|----|----|----|----|----|
| $i$   | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| $p_i$ | 0.01 | 0.01 | 0.03 | 0.04 | 0.04 | 0.05 | 0.13 | 0.18 | 0.24 | 0.27 |    |    |    |    |    |    |    |    |    |
| pred  |      |      |      |      |      |      |      |      |      |      |    |    |    |    |    |    |    |    | –  |
| mark  |      |      |      |      |      |      |      |      |      |      |    |    |    |    |    |    |    |    | –  |

- Run Huffman's algorithm on this input by filling in the table.
- Draw the corresponding optimal coding tree  $T$ . What is the cost of  $T$ ?
- Decode "001101100010" and encode "HEAD".

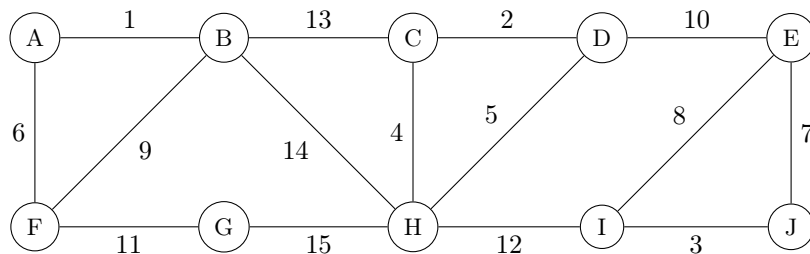
### Exercise 2 (Edge Updates in Minimum Spanning Trees)

Let  $G$  be a weighted graph,  $T$  be a minimum spanning tree for  $G$ , and  $e$  be an edge in  $T$ . Consider the following scenarios.

- We decrease the weight of  $e$ . Show that  $T$  remains a minimum spanning tree for  $G$ .
- We increase the weight of  $e$ . Show that  $T$  does not necessarily remain a minimum spanning tree.

### Exercise 3 (Minimum Spanning Trees – Jarník/Prim and Kruskal's Algorithm)

Consider the following undirected weighted graph  $G = (V, E, c)$ :



- Run the **Jarník/Prim algorithm** on  $G$  with start node A and calculate the weight of the resulting minimum spanning tree  $T$ .
- Run **Kruskal's algorithm** on  $G$  and calculate the weight of the resulting minimum spanning tree  $T$ .